

EXPERIMENT-8

Implement Linear Regression and predict the output for new data.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score

print("="*70)
print("      LINEAR REGRESSION")
print("="*70)

uploaded=files.upload()

data=pd.read_csv("Salary_Data.csv")

print("\nDataset")
print(data)

X=data[['YearsExperience']]
y=data['Salary']

X_train,X_test,y_train,y_test=train_test_split(
X,y,test_size=0.2,random_state=42)

model=LinearRegression()
```

```
print("\nTraining Linear Regression Model...")
```

```
model.fit(X_train,y_train)
```

```
prediction=model.predict(X_test)
```

```
print("\nActual Salary")
```

```
print(y_test.values)
```

```
print("\nPredicted Salary")
```

```
print(prediction)
```

```
print("\nR2 Score")
```

```
print(round(r2_score(y_test,prediction),3))
```

```
print("\nMean Squared Error")
```

```
print(round(mean_squared_error(y_test,prediction),2))
```

```
sample=[[5]]
```

```
print("\nPrediction for 5 Years Experience")
```

```
print("Estimated Salary =",
```

```
round(model.predict(sample)[0],2))
```

Sample Input:

```
... =====  
                        LINEAR REGRESSION  
=====
```

Salary_Data.csv

Salary_Data.csv(text/csv) - 454 bytes, last modified: 8/5/2026 - 100% done
Saving Salary_Data.csv to Salary_Data.csv

Sample Output:

Dataset			
		YearsExperience	Salary
...	0	1.1	39343.0
	1	1.3	46205.0
	2	1.5	37731.0
	3	2.0	43525.0
	4	2.2	39891.0
	5	2.9	56642.0
	6	3.0	60150.0
	7	3.2	54445.0
	8	3.2	64445.0
	9	3.7	57189.0
	10	3.9	63218.0
	11	4.0	55794.0
	12	4.0	56957.0
	13	4.1	57081.0
	14	4.5	61111.0
	15	4.9	67938.0
	16	5.1	66029.0
	17	5.3	83088.0
	18	5.9	81363.0
	19	6.0	93940.0
	20	6.8	91738.0
	21	7.1	98273.0
	22	7.9	101302.0
	23	8.2	113812.0
	24	8.7	109431.0
	25	9.0	105582.0
	26	9.5	116969.0
	27	9.6	112635.0
	28	10.3	122391.0
	29	10.5	121872.0

Training Linear Regression Model...

Actual Salary

[112635. 67938. 113812. 83088. 64445. 57189.]

Predicted Salary

[115790.21011287 71498.27809463 102596.86866063 75267.80422384
55477.79204548 60189.69970699]

R2 Score

0.902

Mean Squared Error

49830096.86

Prediction for 5 Years Experience

Estimated Salary = 72440.66